

Working Principle

The Smart Waste Management System operates as follows:

1. Raspberry Pi Pico W initializes after power-up.
2. The HC-SR04 ultrasonic sensor emits ultrasonic waves.
3. The waves reflect from the garbage surface.
4. The echo time is measured.
5. Raspberry Pi Pico calculates the distance.
6. The garbage fill percentage is calculated.
7. The data is transmitted to the ThingSpeak cloud platform using Wi-Fi.
8. The dashboard displays real-time waste level.
9. If the fill percentage exceeds 80%, the system displays a "Dustbin Full" status.
10. The process repeats continuously.

The complete system is simulated using the Wokwi online simulator.